



Varendra University
Department of Computer Science and Engineering



PROJECT

Course Code : CSE 3102
Course Title : Computer Graphics Lab
Project Title : ATM Interface Using Graphics
Submission Date :

Submitted By	Submitted To
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1. Introduction

An Automated Teller Machine (ATM) is a computerized system that allows users to perform banking transactions without visiting a bank branch. Modern ATMs use graphical interfaces to provide easy interaction between the user and the machine.

This project focuses on designing a graphical ATM interface that simulates real ATM operations such as PIN entry, cash withdrawal, balance inquiry, and deposit transactions. The project uses graphics, buttons, menus, and interactive screens to create a user-friendly banking application.

The purpose of this project is to understand graphical programming concepts and event-driven application development.

2. Problem Statement

Existing Problem

Traditional command-line banking programs are difficult for users to interact with because they lack visual elements and proper navigation. Users may find text-based systems confusing and less attractive.

Problem Addressed by the Project

This project provides a graphical ATM interface that improves usability through:

- Interactive buttons and menus
- Secure PIN entry system
- Easy navigation between banking options
- Real-time transaction simulation
- Better user experience using graphics

3. Objectives

The main objectives of this project are:

1. To create a graphical ATM interface.
2. To simulate ATM banking operations.
3. To implement secure PIN authentication.

4. To provide menu-driven transaction options.
5. To develop an easy-to-use and visually attractive application.
6. To understand graphics programming and GUI development.

4. Algorithms Used

Algorithm for ATM Login

1. Start the program.
2. Display ATM welcome screen.
3. Ask the user to enter PIN.
4. Check whether the PIN is correct.
5. If PIN is correct: Open the main menu. Else: Display "Invalid PIN" and allow retry.

Algorithm for Withdrawal

6. Select "Withdraw Money".
7. Enter withdrawal amount.
8. Check available balance.
9. If balance is sufficient: Deduct amount and display success message. Else: Display "Insufficient Balance".

Algorithm for Deposit

10. Select "Deposit Money".
11. Enter deposit amount.
12. Add amount to balance.
13. Display updated balance.

Algorithm for Balance Inquiry

14. Select "Check Balance".
15. Retrieve current balance.

16. Display balance on screen.

5. Implementation

Programming Language Used

- C++ with OpenGL / GLUT Graphics Library

Graphics Libraries / Tools

- windows.h - Windows API
- GL/glut.h - OpenGL Utility Toolkit
- GL/glu.h - OpenGL Utility Library

Main Components of the System

1. Login Screen

- PIN entry field with asterisk masking
- Numeric keypad (0-9)
- CLEAR, CANCEL, ENTER buttons

2. Main Menu

- FAST CASH
- WITHDRAW
- BALANCE
- LOGOUT

3. Transaction Screens

- Withdraw 5000 Tk confirmation screen
- Success screen with MAIN MENU button

4. Graphics Features

- Colored buttons and rectangles using OpenGL
- Text rendering with GLUT bitmap fonts
- Mouse click detection for interaction
- Screen transitions via enum state machine

Code:

```
#include <windows.h>
#include <GL/glut.h>
#include <GL/glu.h>
#include <iostream>
#include <string>
#include <cstdio>
#include <cstdlib>

using namespace std;

// ===== GLOBAL VARIABLES

enum Screen {
    PIN_SCREEN,
    MENU_SCREEN,
    WITHDRAW_SCREEN,
    SUCCESS_SCREEN
};

Screen currentScreen = PIN_SCREEN;

string pinInput = "";
int accountBalance = 50000;
string statusMessage = "";

// ===== TEXT FUNCTION =====

void drawText(float x, float y, const char* text, void* font) {
    glRasterPos2f(x, y);

    while (*text) {
        glutBitmapCharacter(font, *text);
        text++;
    }
}

// ===== RECTANGLE FUNCTION =====

void drawRect(int x1, int y1, int x2, int y2,
              float r, float g, float b) {
    glColor3f(r, g, b);
    glBegin(GL_QUADS);
```

```

        glVertex2i(x1, y1);
        glVertex2i(x2, y1);
        glVertex2i(x2, y2);
        glVertex2i(x1, y2);

    glEnd();
}

// ===== BUTTON FUNCTION =====

void drawButton(int x1, int y1, int x2, int y2,
                const char* text,
                float r, float g, float b) {

    drawRect(x1, y1, x2, y2, r, g, b);

    glColor3f(1.0f, 1.0f, 1.0f);

    drawText(x1 + 15, y1 + 18,
             text,
             GLUT_BITMAP_HELVETICA_18);
}

// ===== CLICK DETECTION =====

bool isClicked(int mx, int my,
               int x1, int y1,
               int x2, int y2) {

    return (mx >= x1 &&
            mx <= x2 &&
            my >= y1 &&
            my <= y2);
}

// ===== PIN SCREEN =====

void drawPinScreen() {

    glColor3f(1.0f, 1.0f, 0.0f);

    drawText(220, 450,
             "Please Enter Your 4-Digit PIN",
             GLUT_BITMAP_HELVETICA_18);

    // PIN BOXES
    for (int i = 0; i < 4; i++) {

        drawRect(260 + (i * 70),
                 340,
                 310 + (i * 70),
                 390,
                 1.0f, 1.0f, 1.0f);

        if (i < (int)pinInput.length()) {

            glColor3f(0.0f, 0.0f, 0.0f);

            drawText(278 + (i * 70),
                     355,
                     "*",
                     GLUT_BITMAP_TIMES_ROMAN_24);
        }
    }
}

```

```

    }

    // NUMBER BUTTONS

    drawButton(250, 230, 320, 280, "1", 0, 0, 0.8);
    drawButton(350, 230, 420, 280, "2", 0, 0, 0.8);
    drawButton(450, 230, 520, 280, "3", 0, 0, 0.8);

    drawButton(250, 160, 320, 210, "4", 0, 0, 0.8);
    drawButton(350, 160, 420, 210, "5", 0, 0, 0.8);
    drawButton(450, 160, 520, 210, "6", 0, 0, 0.8);

    drawButton(250, 90, 320, 140, "7", 0, 0, 0.8);
    drawButton(350, 90, 420, 140, "8", 0, 0, 0.8);
    drawButton(450, 90, 520, 140, "9", 0, 0, 0.8);

    drawButton(350, 20, 420, 70, "0", 0, 0, 0.8);

    // ACTION BUTTONS

    drawButton(560, 230, 700, 280,
               "CLEAR",
               0.8, 0, 0);

    drawButton(560, 160, 700, 210,
               "CANCEL",
               0.8, 0.5, 0);

    drawButton(560, 90, 700, 140,
               "ENTER",
               0, 0.6, 0);

    // STATUS MESSAGE

    glColor3f(1.0f, 0.0f, 0.0f);

    drawText(260,
             410,
             statusMessage.c_str(),
             GLUT_BITMAP_HELVETICA_18);
}

// ===== MENU SCREEN =====

void drawMenuScreen() {
    glColor3f(1, 1, 1);

    drawText(280, 450,
             "Select Your Transaction",
             GLUT_BITMAP_HELVETICA_18);

    drawButton(150, 280, 350, 340,
               "FAST CASH",
               0, 0.5, 0.5);

    drawButton(450, 280, 650, 340,
               "WITHDRAW",
               0, 0.5, 0.5);

    drawButton(150, 180, 350, 240,
               "BALANCE",
               0, 0.5, 0.5);
}

```

```

        drawButton(450, 180, 650, 240,
                    "LOGOUT",
                    0.8, 0, 0);

// BALANCE DISPLAY

char balStr[100];

sprintf(balStr,
        "Current Balance: %d Tk",
        accountBalance);

glColor3f(0, 1, 0);

drawText(250,
        100,
        balStr,
        GLUT_BITMAP_HELVETICA_18);
}

// ===== WITHDRAW SCREEN =====

void drawWithdrawScreen() {

    glColor3f(1, 1, 1);

    drawText(180,
            380,
            "Withdraw 5000 Tk ?",
            GLUT_BITMAP_HELVETICA_18);

    drawButton(250, 240, 380, 300,
                "YES",
                0, 0.6, 0);

    drawButton(420, 240, 550, 300,
                "NO",
                0.8, 0, 0);
}

// ===== SUCCESS SCREEN =====

void drawSuccessScreen() {

    glColor3f(0, 1, 0);

    drawText(240,
            380,
            "TRANSACTION SUCCESSFUL!",
            GLUT_BITMAP_HELVETICA_18);

    glColor3f(1, 1, 1);

    drawText(250,
            330,
            "Please collect your cash.",
            GLUT_BITMAP_HELVETICA_18);

    drawButton(320, 200, 480, 260,
                "MAIN MENU",
                0, 0, 0.8);
}

```

```

// ===== DISPLAY FUNCTION =====

void display() {
    glClear(GL_COLOR_BUFFER_BIT);

    // BACKGROUND

    drawRect(0, 0, 800, 600,
             0.15f, 0.15f, 0.15f);

    // HEADER

    drawRect(0, 520, 800, 600,
             0.0f, 0.3f, 0.6f);

    glColor3f(1, 1, 1);

    drawText(250,
             550,
             "NATIONAL BANK ATM",
             GLUT_BITMAP_TIMES_ROMAN_24);

    // SCREEN CONTROL

    switch (currentScreen) {
        case PIN_SCREEN:
            drawPinScreen();
            break;

        case MENU_SCREEN:
            drawMenuScreen();
            break;

        case WITHDRAW_SCREEN:
            drawWithdrawScreen();
            break;

        case SUCCESS_SCREEN:
            drawSuccessScreen();
            break;
    }

    glutSwapBuffers();
}

// ===== MOUSE FUNCTION =====

void mouseClicked(int button, int state, int x, int y) {
    if (button == GLUT_LEFT_BUTTON &&
        state == GLUT_DOWN) {

        int mx = x;
        int my = 600 - y;

        // ===== PIN SCREEN =====

        if (currentScreen == PIN_SCREEN) {

```

```

        if (isClicked(mx, my, 250, 230, 320, 280) &&
pinInput.length() < 4)
            pinInput += "1";

        else if (isClicked(mx, my, 350, 230, 420, 280) &&
pinInput.length() < 4)
            pinInput += "2";

        else if (isClicked(mx, my, 450, 230, 520, 280) &&
pinInput.length() < 4)
            pinInput += "3";

        else if (isClicked(mx, my, 250, 160, 320, 210) &&
pinInput.length() < 4)
            pinInput += "4";

        else if (isClicked(mx, my, 350, 160, 420, 210) &&
pinInput.length() < 4)
            pinInput += "5";

        else if (isClicked(mx, my, 450, 160, 520, 210) &&
pinInput.length() < 4)
            pinInput += "6";

        else if (isClicked(mx, my, 250, 90, 320, 140) &&
pinInput.length() < 4)
            pinInput += "7";

        else if (isClicked(mx, my, 350, 90, 420, 140) &&
pinInput.length() < 4)
            pinInput += "8";

        else if (isClicked(mx, my, 450, 90, 520, 140) &&
pinInput.length() < 4)
            pinInput += "9";

        else if (isClicked(mx, my, 350, 20, 420, 70) &&
pinInput.length() < 4)
            pinInput += "0";

        // CLEAR

        else if (isClicked(mx, my, 560, 230, 700, 280)) {
            pinInput = "";
            statusMessage = "";
        }

        // CANCEL

        else if (isClicked(mx, my, 560, 160, 700, 210)) {
            exit(0);
        }

        // ENTER

        else if (isClicked(mx, my, 560, 90, 700, 140)) {
            if (pinInput == "1234") {
                currentScreen = MENU_SCREEN;
                statusMessage = "";
            }
        }

```

```

    } else {
        statusMessage = "Wrong PIN! Use 1234";
        pinInput = "";
    }
}

// ===== MENU SCREEN =====
else if (currentScreen == MENU_SCREEN) {
    if (isClicked(mx, my, 150, 280, 350, 340)) {
        currentScreen = WITHDRAW_SCREEN;
    }
    else if (isClicked(mx, my, 450, 180, 650, 240)) {
        pinInput = "";
        currentScreen = PIN_SCREEN;
    }
}

// ===== WITHDRAW SCREEN =====
else if (currentScreen == WITHDRAW_SCREEN) {
    if (isClicked(mx, my, 250, 240, 380, 300)) {
        if (accountBalance >= 5000) {
            accountBalance -= 5000;
            currentScreen = SUCCESS_SCREEN;
        }
    }
    else if (isClicked(mx, my, 420, 240, 550, 300)) {
        currentScreen = MENU_SCREEN;
    }
}

// ===== SUCCESS SCREEN
else if (currentScreen == SUCCESS_SCREEN) {
    if (isClicked(mx, my, 320, 200, 480, 260)) {
        currentScreen = MENU_SCREEN;
    }
}

glutPostRedisplay();
}

// ===== RESHAPE FUNCTION =====
void reshape(int w, int h) {
    glViewport(0, 0, w, h);
}

```

```

    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();
    gluOrtho2D(0, 800, 0, 600);
    glMatrixMode(GL_MODELVIEW);
}

// ===== MAIN FUNCTION =====

int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB);
    glutInitWindowSize(800, 600);
    glutInitWindowPosition(100, 100);
    glutCreateWindow("ATM Machine");
    glClearColor(0, 0, 0, 1);
    glutDisplayFunc(display);
    glutReshapeFunc(reshape);
    glutMouseFunc(mouseClick);
    glutMainLoop();
    return 0;
}

```

6. Results / Outputs

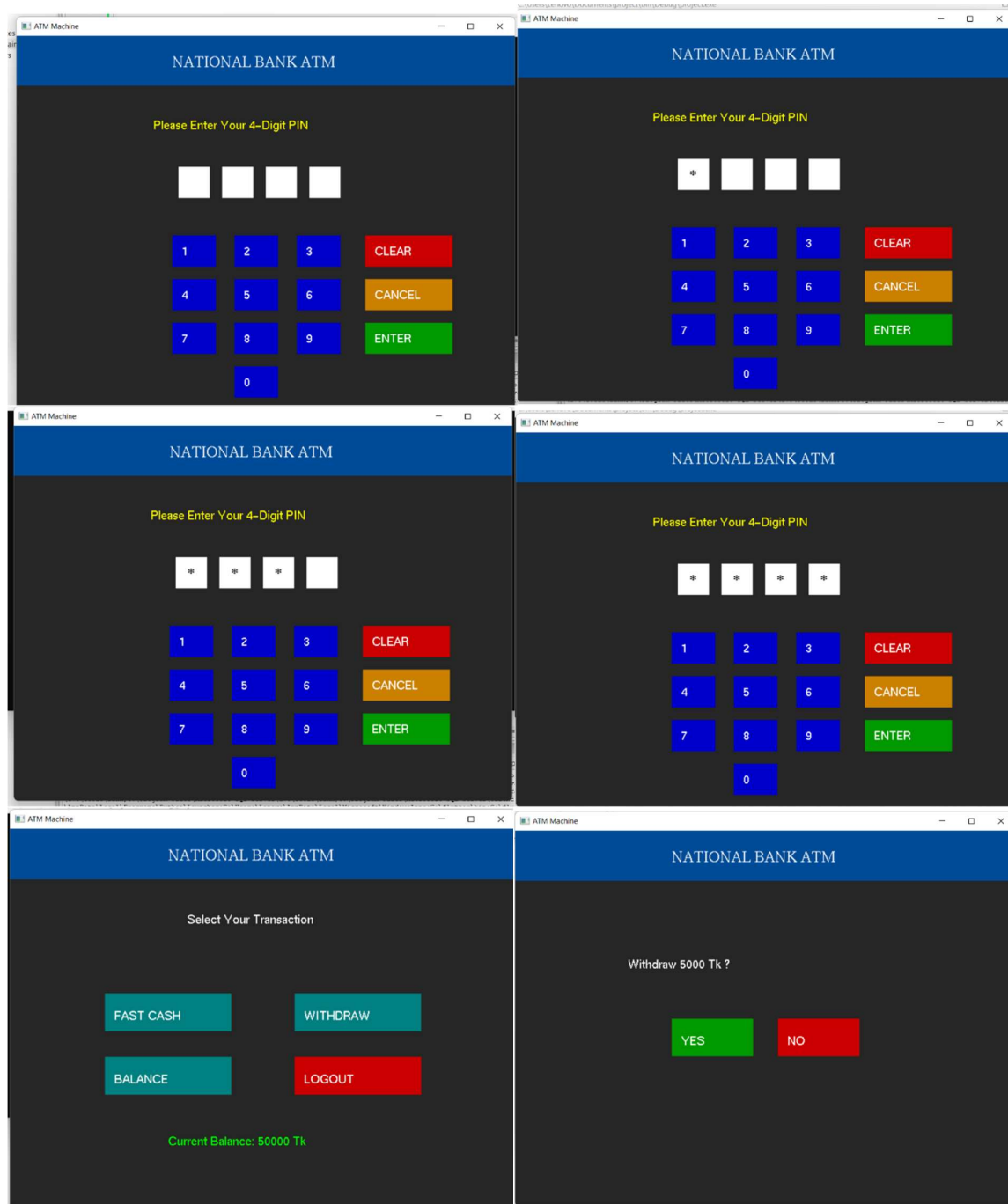
The ATM graphical interface successfully performs:

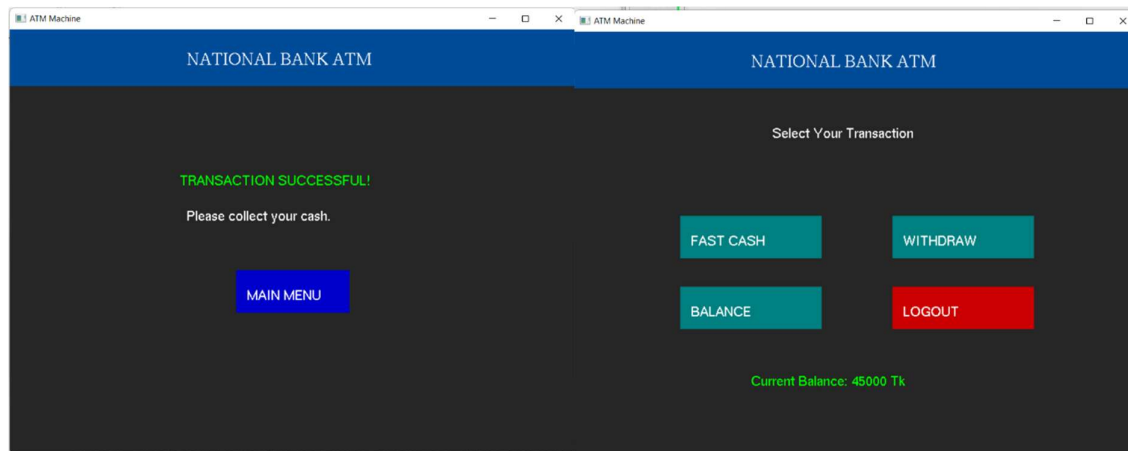
- User authentication using PIN (correct PIN: 1234)
- Cash withdrawal of 5000 Tk with balance deduction
- Menu navigation through clickable buttons
- Error message display for wrong PIN

Sample Outputs

- Welcome screen with PIN entry displayed successfully
- Correct PIN (1234) opens the main transaction menu

- Incorrect PIN shows "Wrong PIN! Use 1234" error message
- Balance updates from 50000 Tk to 45000 Tk after withdrawal





7. Conclusion

The ATM Interface Using Graphics project successfully demonstrates the use of graphical programming in developing a banking application. The project improves user interaction through visual menus and buttons while simulating real ATM operations.

This project helped in understanding:

- GUI design using OpenGL and GLUT
- Event handling with mouse callbacks
- Graphics programming with 2D rendering
- Banking transaction logic and state management

Future improvements may include:

- Database connectivity for real account data
- Fingerprint authentication
- Online banking integration
- Multi-user support